The Assessment Burden in Competency-Based Medical Education: How Programs Are Adapting

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Abstract

Residents and faculty have described a burden of assessment related to the implementation of competency-based medical education (CBME), which may undermine its benefits. Although this concerning signal has been identified, little has been done to identify adaptations to address this problem. Grounded in an analysis of an early Canadian pan-institutional CBME adopter's experience, this article describes postgraduate programs' adaptations related to the challenges of assessment in CBME. From June 2019–September 2022, 8 residency programs underwent a standardized Rapid Evaluation guided by the Core Components Framework (CCF). Sixty interviews and 18 focus groups were held with invested partners. Transcripts were analyzed abductively using CCF, and ideal implementation was compared with enacted implementation. These findings were then shared back with program leaders, adaptations were subsequently developed, and technical reports were generated for each program. Researchers reviewed the technical reports to identify themes related to the burden of assessment with a subsequent focus on identifying adaptations across programs. Three themes were identified: (1) disparate mental models of assessment processes in CBME, (2) challenges in workplace-based assessment processes, and (3) challenges in performance review and decision making. Theme 1 included entrustment interpretation and lack of shared mindset for performance standards. Adaptations included revising entrustment scales, faculty development, and formalizing resident membership. Theme 2 involved direct observation, timeliness of assessment completion, and feedback quality. Adaptations included alternative assessment strategies beyond entrustable professional activity forms and proactive assessment planning. Theme 3 related to resident data monitoring and competence committee decision making. Adaptations included adding resident representatives to the competence committee and assessment platform enhancements. These adaptations represent responses to the concerning signal of significant burden of assessment within CBME being experienced broadly. The authors hope other programs may learn from their institution's experience and navigate the CBME-related assessment burden their invested partners may be facing.

Competency-based medical education (CBME) is in various stages of implementation across the globe. CBME can be successful and continue as a viable educational model only if implementation occurs as intended with its desired outcomes achieved. Calls have been made for evaluation of CBME's effectiveness[1] (ultimately, improved patient care[2]), which will involve long-term evaluation strategies. A critical first step before outcome evaluation will be ensuring that CBME was introduced in the manner intended, referred to as fidelity of implementation.[3] To this end, evaluation strategies to understand the fidelity of varying aspects of CBME implementation to date have used a variety of approaches and frameworks, including realist evaluation,[4] Rapid Evaluation,[5] logic models,[1] contribution analysis,[6] process and outcomes evaluation,[7] the CBME Core Components Framework (CCF),[8] and many others. These studies have noted a variety of successes and challenges in implementation, but one of the clearest early struggles has been the identification of a major burden of assessment widely experienced by both residents and faculty.[9]

Workplace-based assessment is central to CBME and has 2 main purposes.[10] The first is assessment for learning: low-stakes and formative assessment to support learning, the ongoing development of self-assessment skills, and documentation of low-stakes individual data points. The second is assessment of learning, through the aggregation of those data points and as a critical component in a program of assessment to identify patterns of performance and learning trajectories to make high-stakes and summative decisions about residents' progress and inform the development of learning plans. However, tension in this dual purposing of assessment has been identified as an issue.[11,12] Concerns include the fact that formative assessment, instead of being used to facilitate learning and enhance clinical learning encounters, has been reduced to a time-consuming tick-box exercise, essentially limiting the focus and value of assessment.[13,14] Others[13] have questioned whether residents truly view low-stakes assessments in the nonsummative light that was intended. Moreover, residents have been observed to selectively seek formal assessment only when confident of success.[15,16] These performance-focused strategies, as well as the burden of assessment in general, have the potential

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Acad Med. 2023;98:1261–1267.
First published online June 21, 2023
doi: 10.1097/ACM.0000000000005305

Supplemental digital content for this article is available at http://links.lww.com/ACADMED/B438.

to subvert the intended developmental focus and growth mindset of CBME.[17] Previous work, including at the authors' institution,[14] has established the presence of these concerns and has identified them as a major threat to the fidelity of CBME implementation and buy-in. Furthermore, the concerns related to assessment in CBME are associated, in theory, with concerns about the deterioration of faculty coaching and feedback efforts and their own burnout as well as with issues related to resident wellness.[9] Despite these significant experienced challenges, there is little in the literature describing potential program-level adaptations that could mitigate these risks.

Queen's University was an early adopter of CBME and implemented it collectively across residency postgraduate training programs in 2017. This unique position allows for a description of the institution's experiences across multiple disciplines over time. As part of this process, there was an intentional, prospective, program-specific evaluation process organized and driven by Postgraduate Medical Education that was based on the CCF that was adaptation focused and aimed to optimize fidelity of local implementation across postgraduate programs.[18] Grounded in an analysis of the program evaluation technical report data derived from this process, the aims of this study were to (1) identify themes related to the perceived burden of assessment across 8 postgraduate training programs and (2) describe program-level proposed and/or enacted adaptations (both common and unique to individual programs) to address these challenges. The authors' hope is that other programs and institutions transitioning to CBME might be able to see themselves in the data and learn from this experience, improving their own transition to CBME, particularly as it relates to the experienced burden of assessment.

Method and Participants

This study was conducted at Queen's University and its affiliated hospital, Kingston Health Sciences Centre, in Kingston, Ontario, Canada. Queen's University was granted Fundamental Innovations in Residency Education status by the Royal College of Physicians and Surgeons of Canada to transition its residency programs to CBME curricula ahead of the national Canadian Competency by Design schedule. Institutional program leader biannual workshops were initiated in January 2015 and introduced the rationale for CBME, an overview of the proposed curriculum alignment process, the development of stage-specific entrustable professional activities (EPAs), required training experiences, programs of assessment, and preparation for launch. In July 2017, approximately 70 incoming residents joined newly developed CBME-focused specialty residency programs at Queen's University. This cross-discipline study involved the analysis of technical reports generated through individual program evaluation studies in 8 specialty programs: diagnostic radiology, gastroenterology, general surgery, obstetrics and gynecology, orthopedic surgery, pediatrics, physical medicine and rehabilitation, and surgical foundations. Data analysis was completed between June 2019 and September 2022. These programs were the first to complete this standardized program evaluation approach. Institutional research ethics board approval was obtained, and all participants provided informed consent.

Conceptual framework and evaluation methods

The overall evaluation was guided by the CBME CCF[8] because this framework was believed to be the most relevant for the authors' context and was endorsed by the International Competency-Based Medical Education Collaborators.[19] The CCF was developed to support program evaluations of CBME implementation and includes 5 core components of a training program: outcome competencies, sequenced progression, tailored learning experiences, competency-focused instruction, and programmatic assessment. With the use of the CCF, this evaluation followed the Rapid Evaluation of CBME protocol 5 for each training program in a standardized fashion as described by Hall et al.[5] The key steps were as follows: (1) identification of the implementation as intended (anticipated behaviors and outcomes organized by CCF components), (2) description of programs' actual experience and outcomes related to CBME, and (3) proposed program adaptions based on an analysis of the ideal vs enacted outcomes for each specialty program.

Data collection and analysis

Data were collected in 3 phases. Phase 1 of data collection occurred at the individual program level. Through the above Rapid Evaluation approach, ideal implementation, which was documented a priori, was compared with enacted implementation, measured via interviews and focus groups with different invested partner groups, including program leaders (program directors, competence committee [CC] members, academic advisers, educational consultants, program assistants, and CBME leads), residents, and faculty. This comparison between ideal and enacted implementation formed the basis for initiating a discussion between institutional postgraduate educational leaders and program leaders around future proposed adaptations at the program level.

Individual program-level detailed technical reports were generated, and interested readers are encouraged to review the article by Chung et al,[20] which provides a detailed description of the data collection and analysis process for 1 of the programs included in this study. In total, across the 8 programs, there were 18 focus groups and 60 individual interviews, involving a total of 51 frontline faculty (1–10 per program), 63 residents (6–11 per program), and 61 program leaders (4–15 per program). Interviews and focus groups were selected as data collection sources given the desire to elicit discussion and elaboration about experiences with CBME implementation. Sample interview and focus group questions are available in Supplemental Digital Appendix 1 at http://links.lww.com/ACADMED/B438. Facilitators for the interviews and focus groups had experience in conducting education scholarship. However, they underwent additional training delivered by health professions education scholars with expertise in conducting qualitative studies (H.B. and N.D.). The training included observing H.B. or N.D. conducting an interview and a focus group followed by the facilitators conducting an interview or focus group and receiving feedback from H.B. and N.D. Of note, faculty often held multiple roles; as such, there were 61 unique faculty members who were interviewed during the study. Any differences identified between invested partner groups were highlighted within the technical reports because program leaders appreciated knowing whether findings were specific to a single partner group or identified across partner groups.

In phase 2, N.D. and H.B. met to review the technical reports from the 8 programs with a specific focus on comparing subthemes and themes related to the burden of assessment. Burden of assessment was chosen as the focus of this phase because it was referenced consistently and often by all invested partners and across all programs during phase 1. This analysis focused on identifying any instance associated with the burden of assessment within each of the CCF elements across the 8 technical reports. Once all instances were identified, a working group composed of a subset of the research team (H.B., D.J.D., L.M., and N.D.) met to group them into related subthemes and identify the overarching themes. The full research team then met to discuss the findings and reached consensus on both the subthemes and themes.

Phase 3 focused on identifying adaptations to each burden of assessment common across the 8 programs and those unique to individual programs. Adaptations were strategies that program leaders, in collaboration with the evaluation team, identified for use within their program as a means of addressing the gaps and recommendations described through the interviews and focus groups. Similar adaptations were consolidated and organized within the identified themes related to burden of assessment from stage 2. Unique adaptations were labeled as such. The full research team met to discuss the adaptations and their organization until consensus was reached for all adaptations within each theme, common and unique.

Research team reflexivity

The team included members from different backgrounds, including education (H.B., N.D., and L.M.), family medicine (K.W.S.), and emergency medicine (A.S., A.K.H., and D.J.D.). They provided different perspectives and encouraged each other to share their thinking and challenge assumptions. All authors have additional research training and extensive experience in conducting health professions education scholarship. This was an iterative process, with all team members maintaining an ongoing dialogue[21,22] to ensure a shared understanding. The authors regularly discussed their interpretations of the results to ensure these interpretations were representative of the data and to better understand how their own experiences shaped their participation in the analysis and writing process. Two researchers (H.B. and N.D.) that led all data collection and analysis are nonclinicians and hold positions outside postgraduate medicine. They had the appropriate context for CBME given their previous scholarship work but had no vested interests in CBME implementation at the institution. This helped to mitigate potential bias associated with collecting data at their institution.

Results

Adaptations were collected across all 8 programs. Each program identified adaptations related to the burden of assessment. We collated adaptations across programs and organized them by the 3 major identified burden of assessment themes. These themes are presented in Table 1 alongside their associated subthemes. A full overview of themes, subthemes, and adaptations are available in Table 2.

Three burden of assessment themes (and associated subthemes) were identified. Disparate mental models of assessment processes in CBME (theme 1) included different entrustment interpretation and lack of shared mindset for performance standards. Adaptations included revising entrustment scales, faculty development, and formalizing resident membership. Challenges in workplace-based assessment processes (theme 2) focused on direct observation, timeliness of assessment completion, and feedback quality. Adaptations included alternative assessment strategies beyond EPA forms and proactive planning for assessments. Challenges in performance review and decision making (theme 3) included resident data monitoring and CC decision making. Adaptations included adding resident representatives to the CC and assessment platform enhancements.

Table 1
Overview of Identified Assessment Burden Themes and Subthemes

Theme	Subtheme
Disparate mental models of assessment processes in CBME (theme 1)	• Variable scale interpretation • Shared performance standards
Challenges in workplace-based assessment processes (theme 2)	• Direct observation • Timeliness of assessment completion • Documentation of performance • Feedback quality
Challenges in performance review and decision making (theme 3)	• Monitoring performance • Competence committee decision making

Discussion

The identified burden of assessment[23–25] in the rollout of CBME is a major barrier to the fidelity of its implementation and is believed to have a negative impact on resident and faculty wellness.[26,27] This has become a dominant theme in the discourse related to CBME uptake. In this study, based on the authors' unique position as pan-institutional early adopters of CBME in postgraduate medicine, 3 main themes related to the burden of assessment were experienced by invested partners. These identified themes span the spectrum of assessment, starting with understanding the purpose of assessment through data collection, interpretation, and progression decision making. These findings are in line with the findings of other work that have identified the unintended burden of assessment within CBME in numerous settings for both faculty and residents.[9] What this work adds to the literature is the identification of both program-specific and cross-program adaptations (proposed and/or enacted) by residency programs related to each of the identified themes. These adaptations were identified by individual programs, factoring in their own contexts and unique characteristics.

Adaptations related to disparate mental models of assessment processes in CBME (theme 1)

We identified several adaptations related to disparate mental models of

Table 2
Overview of Common and Unique Adaptations Linked to Assessment Burden Themes

Assessment burden theme	Common adaptations	Unique adaptations
Disparate mental models of assessment processes in CBME (including subthemes of variable scale interpretation and lack of shared performance standards) (theme 1)	• Revising entrustment scales for ease of use and clarity • Revisiting conceptualizations of stages and EPAs (to address fragmentation within and across stages, progression, and levels of performance within and across EPAs) • Offering faculty development to clarify entrustment and EPAs (e.g., micro-opportunities linked to other meetings, sharing exemplars, and multipronged communication strategies) • Sharing tips-and-tricks emails to support assessment completion (e.g., use of PIN and in-the-moment completion)	• Revising resident resources (e.g., orientation package) • Formalizing meaningful resident representation on RPC through terms of reference specific to eliciting and sharing feedback at RPC • Promoting end-of-block discussions between residents and faculty about CBME that includes a review of entrustment scale • Reviewing different assessments to help residents and faculty understand their unique value and contribution
Challenges in workplace-based assessment processes (including subthemes of direct observation, timeliness of assessment completion, documentation of performance, and feedback quality) (theme 2)	• Encouraging direct observation through dedicated clinical time for faculty • Leveraging assessment strategies beyond EPA forms (e.g., periodic performance assessments) • Providing faculty development specific to enhancing coaching in the moment and feedback quality, encouraging assessment initiation and completion, and periodically reviewing software functionality • Planning and initiating assessments proactively (e.g., distributing case load at beginning of shift according to EPA needs and striving for balance in who initiates assessments)	• Highlighting for faculty and resident assessment information that is most useful to the CC in formulating progress and promotion decisions • Raising awareness about assessment opportunities for residents and faculty through EPA cards in clinic • Clarifying expectations for residents and faculty • Offering incentives to enhance faculty engagement (e.g., competition to see who uses their PIN first and/or most frequently) • Creating CBME report cards for each faculty member, including number, variety, comprehensiveness, and timeliness of completed assessments • Developing prompts to enhance faculty and resident engagement with EPAs (e.g., color-coded ring-cards mapping high-yield and high-value EPAs to clinical environments) • Reminding residents which EPAs are of highest value at the beginning of each block and documenting that in Elentra rotation schedule for easy access • Meeting regularly with residents to discuss CC decisions and identify strategies to maximize their learning using the EPAs • Clarifying expectations and empowering learners to initiate EPA assessments and support completion in the moment
Challenges in performance review and decision making (including subthemes of monitoring performance and competence committee decision making) (theme 3)	• Enhancing reporting functionality in Elentra software and strategically supplementing by creating tracking sheets and generating graphs • Offering professional development opportunities (academic half-days, education rounds, and departmental meetings)	• Enhancing transparency of CC with a resident nominated representative • Hosting CC retreat to review each EPA and develop a shared understanding of EPA achievement, CBME processes, and expectations • Feeding back to specialty residency committee with subsequent changes to EPAs and target numbers • Tracking EPA completion by program leaders and CC every 4 blocks, including reviewing resident progress in relation to other residents

Abbreviations: CBME, competency-based medical education; CC, competence committee; EPA, entrustable professional activity; PIN, personal identification number; RPC, residency program committee.

assessment processes. In the context of this work, mental model refers to a shared understanding of a concept. A common adaptation across several programs was to review and revise workplace-based assessment scales with entrustment anchors to make them more intuitive and contextually relevant. Despite the noted benefits of entrustment anchors over traditional scales, contextual implementation challenges have been identified,[28] prompting a discourse regarding the naming and framing of these scales as well as their retrospective vs prospective nature.[29,30] Incongruence or ambiguity between what is entrustable in the clinical environment and scale anchors can lead to unpredictable individual interpretation of the meaning of the scales, low-quality assessment, and a lack of engagement in the assessment process.[31,32] The cognitive dissonance[33] caused by this ambiguity can increase cognitive load[26] and demotivate faculty in completing quality assessments. Preceptors' mental energy should be focused on providing meaningful feedback, both oral and written. Strategies that decrease cognitive dissonance leave more bandwidth for this important task, which may increase the likelihood of meaningful feedback. Revising assessment scales that intuitively make sense also increases the likelihood of meaningful convergence among users. Given that having multiple assessors for each resident is a key element of workplace-based assessment, confidence that scales are being interpreted similarly across assessors is important.[32] Furthermore, faculty may benefit from additional training related to the provision of quality feedback. All of this would increase confidence that differences in assessment are tied to differences in resident performance rather than differences in assessors' interpretation of the scale.

Faculty and resident development sessions were another commonly proposed adaptation across programs. These sessions serve several purposes, including increasing meaningful convergence about scales and supporting change buy-in if time is also allocated to hear people's concerns and ideas for solutions.[34] Having residents be a part of these sessions has the value of facilitating the cocreation of solutions with residents and increasing their sense of agency. Some programs have also chosen to have faculty in certain rotations or experiences focus on a limited number of EPAs, which serves to decrease their assessment-related cognitive load and builds experience and expertise in those relevant EPAs.

Adaptations to challenges in workplace-based assessment processes (theme 2)

Programs identified numerous adaptations to address challenges in workplace-based assessment processes. The creation of ring-cards of EPAs (small, laminated cue cards connected by a metal ring) was believed to mitigate technological fatigue, avoid needing to remember a personal identification number, and navigate the "clicks" of an electronic pathway. These ring-cards were also believed to be helpful in areas such as the operating room, where computers are not always accessible and scrolling through EPAs on a telephone is not practical. Others have used this approach as well, with important positive outcomes.[35] Related to this, programs proposed arranging these cards by rotation to provide more specific direction to residents about high-yield EPAs for specific rotations. This approach could prevent residents from missing opportunities to acquire assessments on rare EPAs, thereby decreasing residents' assessment anxiety about failure to progress because of a lack of performance data. Assessment must be made easy if it is to be performed in the frequent and intended manner of CBME. If tasks are streamlined, they are more likely to get completed as intended.[36] The simple, no-tech addition of ring-cards breaks down some of the barriers related to recalling EPAs and optimizes assessment opportunities.

Encouraging direct observation with dedicated and protected clinical time was another adaptation identified within this theme. This adaptation emphasizes the importance of this work, frees up the mental space for the preceptor to focus on direct observation and/or give thoughtful feedback, and provides time to complete written feedback. Several medical education thought leaders[37] have advocated that this adaptation may also help with creating dedicated feedback and/or assessment. It may also help with faculty wellness through decreasing demands on faculty time. Faculty initiating documentation of feedback signals to the resident the importance the preceptor places in this work, stops the process of resident cherry picking, and, given the number of times residents indicate that they feel the burden of assessment is all on them, starts to ease the burden on residents. It is important to acknowledge that workplace-based assessment (which requires direct observation as well as oral and written feedback) takes more time to do well and may decrease clinical productivity. There may need to be institutional and/or departmental discussions about the priority of education as a fundamental aim of a teaching institution and a realignment of resources. Intentional system changes and allowances are needed to provide faculty with the time and bandwidth to perform this important task, rather than just telling faculty and residents that they need to "do more" with the time they have.

Incentivizing or motivating faculty engagement in assessment processes was a unique adaptation in response to challenges of residents getting faculty to complete EPA assessments. Tracking down faculty to complete EPA assessments is one of the largest contributors to resident stress.[26] Proposed incentives include providing faculty report cards and related quality assurance reporting that provides feedback to faculty about the quality of their feedback and its timeliness compared with their peers.[38,39]

Adaptations related to challenges in performance review and decision making (theme 3)

Programs identified several adaptations related to challenges reviewing resident performance and associated decision making. One program's related adaptation was to include a resident on the CC to add transparency for residents about the CC decision-making process. Approximately 20% of Canadian CCs[40] have used this adaptation, which helps to mitigate some of the anxiety for residents about CCs.[5,9] A concern raised about this approach has been the need to protect the privacy of resident assessment and progression data. In our experience, a resident representative on the CC has been successful when requested by the residents and when there is deliberate attention to confidentiality and conflict of interest guidance for the CC.

Another commonly identified adaptation was the development of a CC retreat during which each EPA was reviewed to ensure a shared and deep understanding of what demonstrates achievement of each EPA. Such a retreat may allow for more flexibility in CC decision making in which quality and variation in experiences, as well as quantity, are all considered when determining residents' progression and attainment of competence. This more nuanced appreciation of competence decisions may also help address residents' anxiety related to the perception of EPA numbers being the primary driver of progression.

Electronic learning platform issues were mentioned consistently across all invested partners and programs. A key element for the success of CBME implementation is an electronic platform that is nimble and adaptable and displays data to facilitate interpretation by all invested partners.

Importantly, this is an institutional-level adaptation because individual programs are powerless to make meaningful change on this front. When they function well, electronic platforms can nudge invested partners to take ownership of their role in assessments and ease the burden on CCs, program directors, residents, and program assistants. When they do not function well, significant tension arises, invested partners begin to disengage, and fidelity of CBME implementation, as a whole, suffers.

Conclusions

From the start, it was clear that implementing CBME was going to increase the volume of work for residents and faculty.[41] The high-quality assessment that is central to CBME is known to take more time. However, based on the data presented herein and elsewhere, the increased volume of work only partially explains the burden of assessment now being felt.

The additional burden placed on both residents and faculty can be partially explained by considering elements of cognitive load theory,[42] in particular, the constraints of human working memory.[43] When individual residents are responsible for completing extraneous tasks not directly related to their education and/or clinical care (e.g., tracking down faculty members to complete assessment forms), they have less bandwidth for important tasks intrinsic to their learning, such as self-reflection and seeking useful feedback or coaching. The resultant cognitive overload[44] may represent part of the identified burden of assessment being felt. At the same time, cognitive overload related to assessment may impact faculty members juggling numerous competing priorities, resulting in poor-quality feedback for residents as well as their own burnout.

The assessment burden–related proposed adaptations outlined in this article are borne out of the experiences at the authors' institution for the last several years. The authors' intention is for others transitioning to CBME to learn from these experiences and mitigate some of the burden of assessment that invested partners have faced to date. Of note, the goal of this work was not to provide evidence for or against CBME but rather to discuss practical strategies developed from the authors' uniquely broad institutional experience.

This study is limited by the fact that it represents a single center's experience with CBME rollout. That being said, the pan-institutional rollout had the benefit of controlling for within-institutional variables that otherwise would have complicated the analysis. This work is also limited by the reliability of the data in the technical reports. To optimize this, the program evaluation team met with program leaders during the generation phase of the reports to ensure they had an opportunity to provide input and correct any misunderstandings. Other limitations include small numbers in some residency programs, subjective interpretation of findings, and cultural differences between Canada and other areas of the world. Future studies should revisit perceptions of the burden of assessment after the programmatic adaptations described herein have been enacted to see whether they have had the expected positive impact on education and whether this has improved the fidelity of CBME implementation as intended.

In the context of Queen's University's early transition to CBME within postgraduate medicine, assessment burden has been frequently described. This article presents program-level adaptations that address this concerning signal and are meant to be helpful and relevant. The goal is for other programs and institutions to learn from the authors' collective experience and better navigate the CBME-related assessment burden they may be facing.

Acknowledgments: The authors would like to thank all the participants who shared their experiences as part of this evaluation.

Funding/Support: This study was supported by the Queen's University Postgraduate Office.

Other disclosures: None reported.

Ethical approval: This study was approved by the Queen's University and Affiliated Hospitals Health Sciences Research Ethics Board, March 25, 2015, #6015151.

Previous presentations: Preliminary findings related to these data were presented at the 2022 Royal College of Physicians and Surgeons of Canada Program Evaluation Summit, April 12–18, 2023, Quebec City, Quebec, Canada, and the 2023 International Congress on Academic Medicine, June 5, 2023, Montreal, Quebec, Canada.

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